

PRACTICAL 9 : ENTERPRISE NETWORK DEPLOYMENT AND TROUBLESHOOTING

WEEK 9

Objective

Deploy and troubleshoot a complete Cisco-based enterprise network.

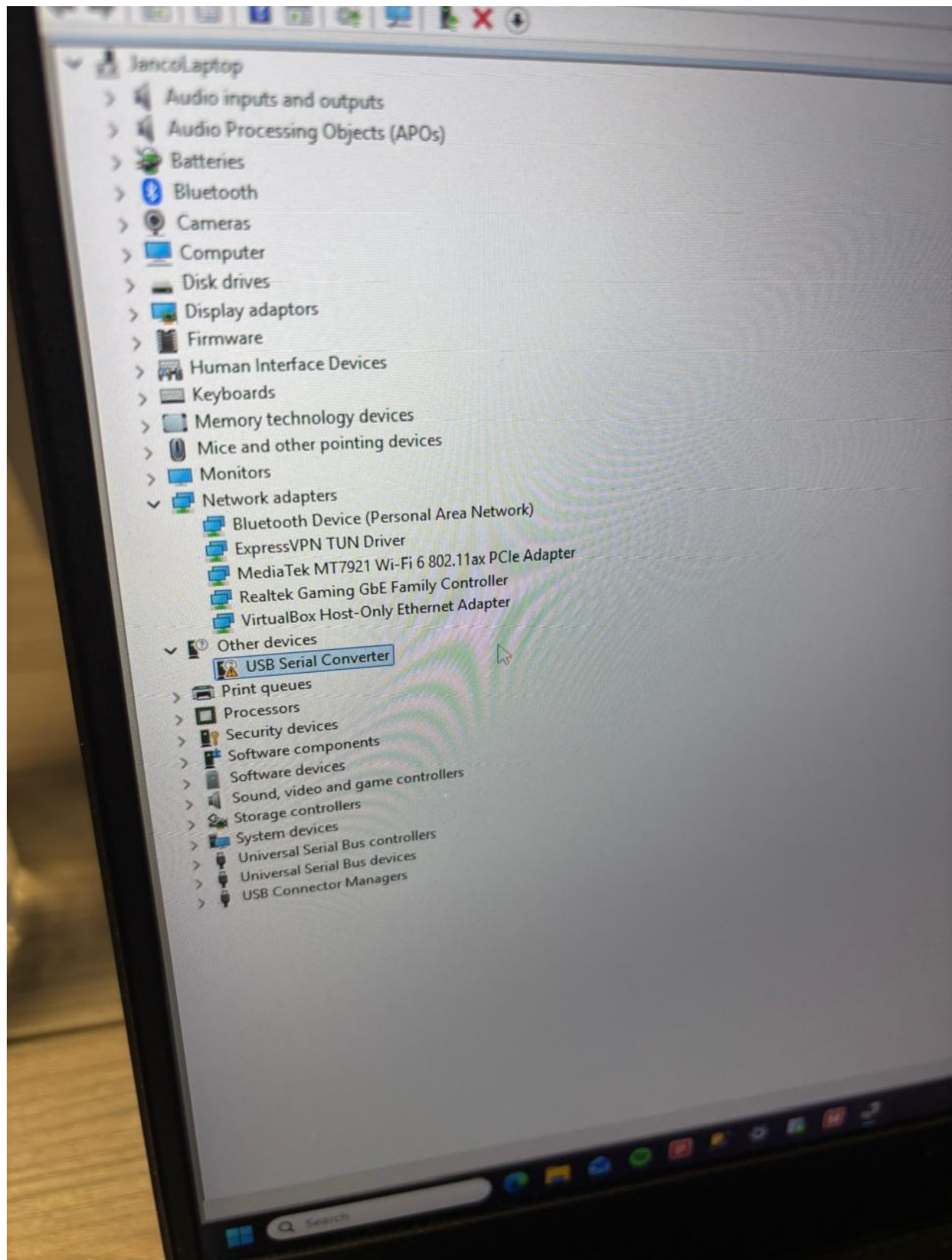
Equipment

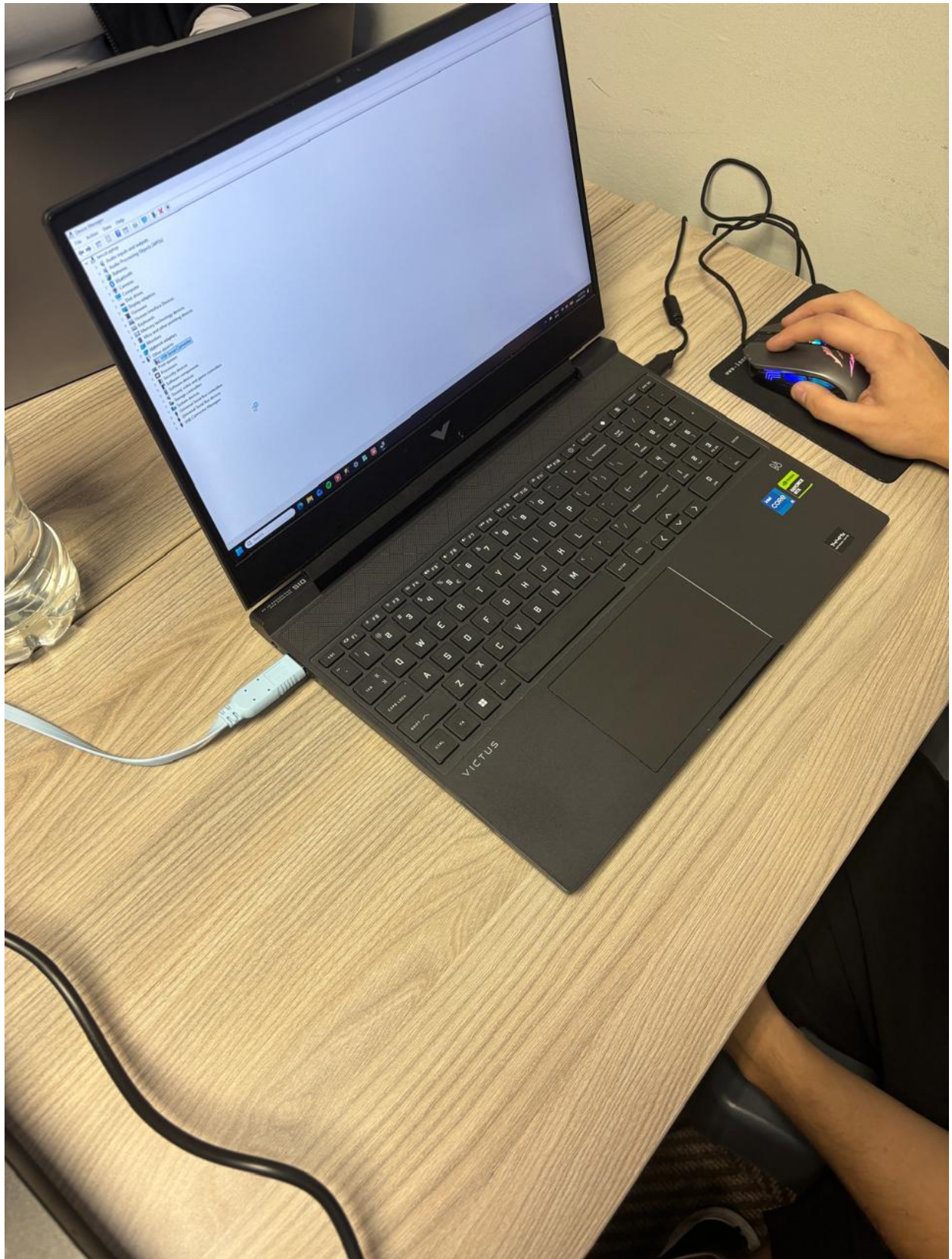
- Cisco Routers
- Cisco Switches
- Switch Cabinets
- Patch Panels

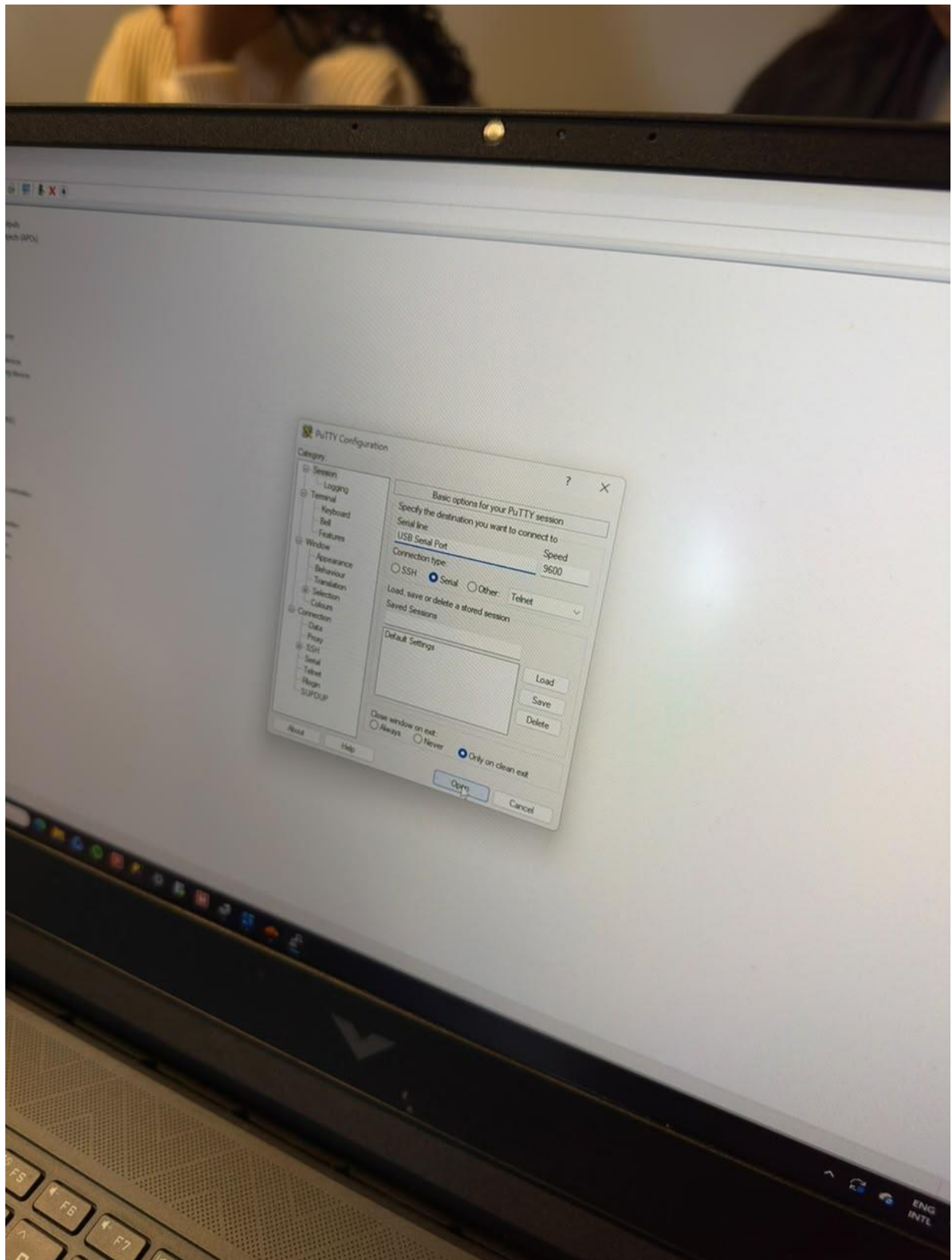
- Printers
- 10 PCs

Activities





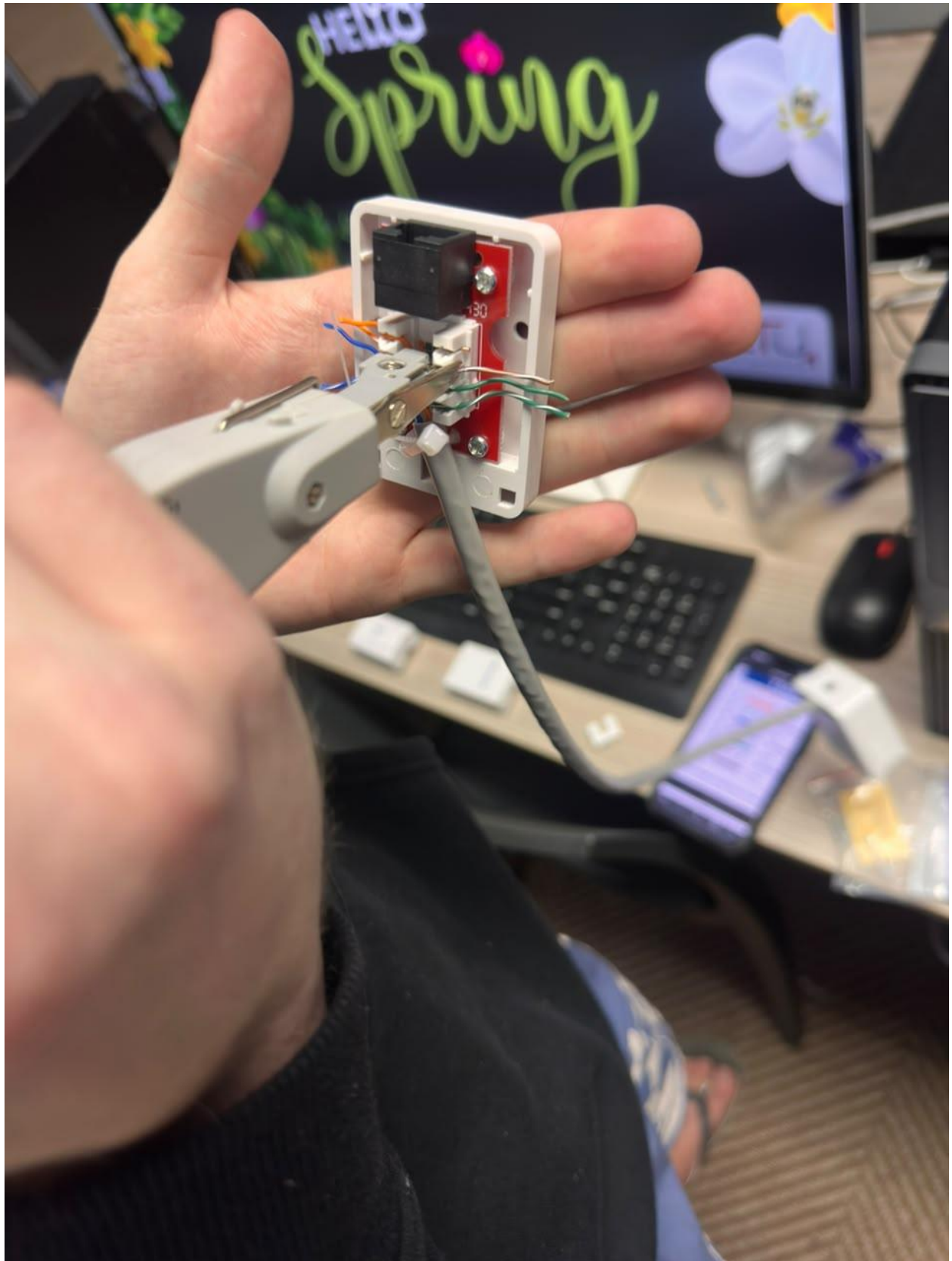






10:57 AM
10/2/2026







PC1 = Fe0/1 ~ 192.168.10.2

PC2 = Fe0/2 ~ 192.168.10.3

PC3 = Fe0/3 ~ 192.168.10.4

PC4 = Fe0/6 ~ 192.168.10.5

PC5 = Fe0/5 ~ 192.168.10.6

Vlan 10

PC6 = Fe0/6 ~ 192.168.20.2

PC7 = Fe0/7 ~ 192.168.20.3

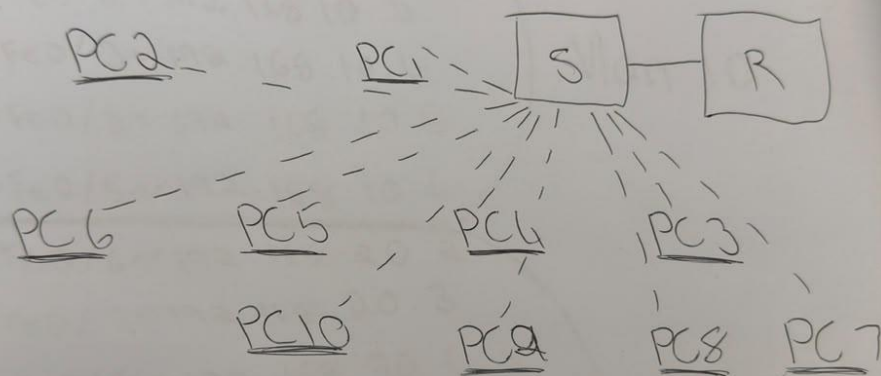
PC8 = Fe0/8 ~ 192.168.20.4

PC9 = Fe0/9 ~ 192.168.20.5

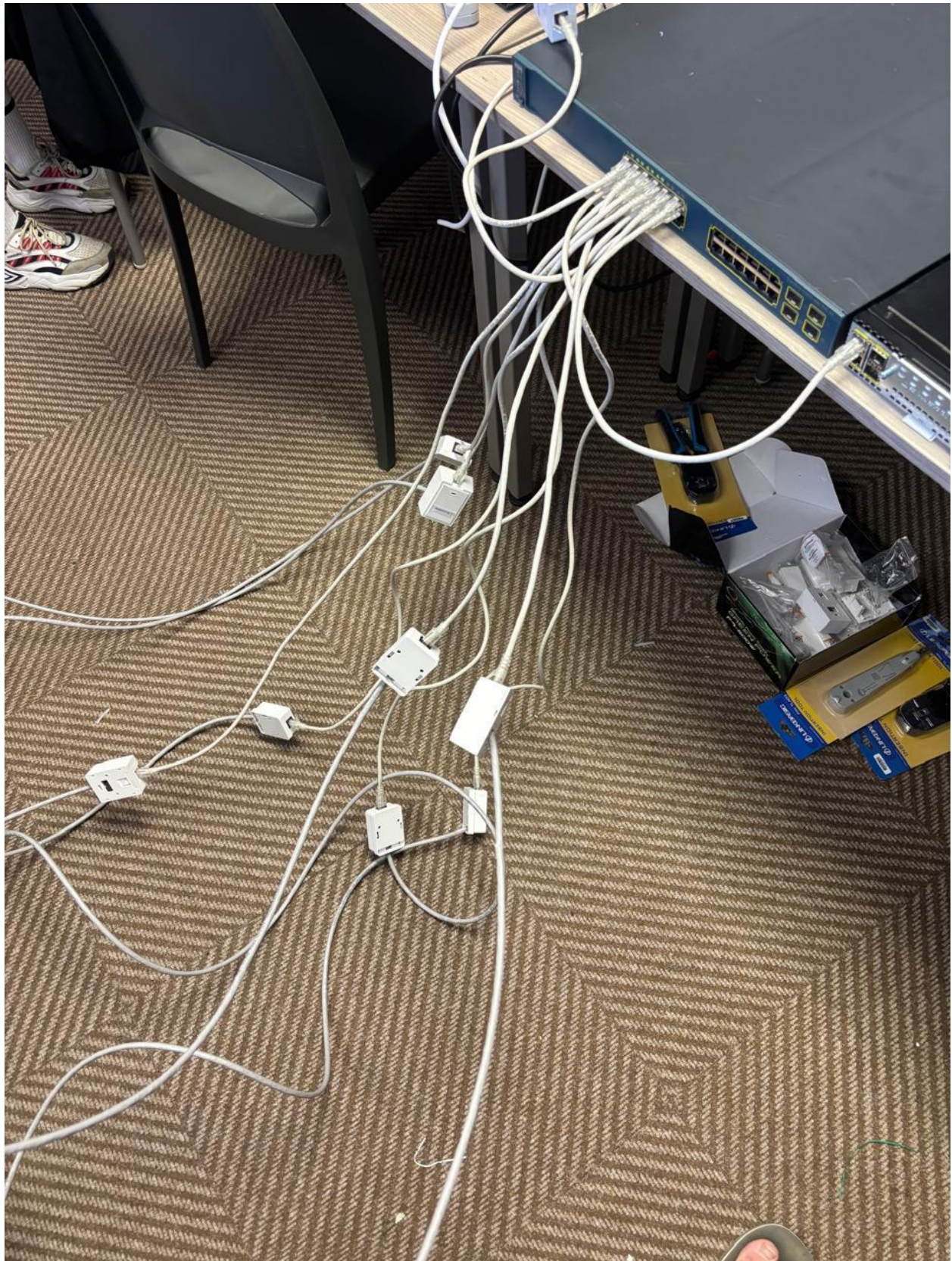
PC10 = Fe0/10 ~ 192.168.20.6

Vlan 20

Topology







1. Install switch cabinet.
2. Install patch panels.
3. Configure switches.
4. Configure routers.
5. Implement VLANs.
6. Configure printers.
7. Test and troubleshoot the network.

Introduced Faults

- Incorrect VLAN assignment
- Incorrect subnet mask
- Incorrect default gateway
- Duplicate IP address
- Faulty patch lead
- Disabled switch port
- Router interface shutdown
- Missing route

Troubleshooting Commands

```
1 show running-config
2 show vlan brief
3 show interfaces
4 show ip route
5 show ip interface brief
6 show mac address-table
7 ping
8 tracert
```

Deliverable

Completed troubleshooting report.

Assessment Rubric (10 Marks)

Criteria	Marks
Fault identification	3
Troubleshooting methodology	3
Correct resolution	2
Professional report	2

Total	10
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PRACTICAL OBSERVATION CHECKLIST

Competency	Yes	No
Uses tools safely	✓	☐
Uses crimping tools correctly	✓	☐
Uses punchdown tools correctly	✓	☐
Applies T568B correctly	✓	☐
Configures Cisco devices correctly	✓	☐
Demonstrates troubleshooting skills	✓	☐
Maintains clean workstation	✓	☐
Demonstrates teamwork	✓	☐
Produces accurate documentation	✓	☐
